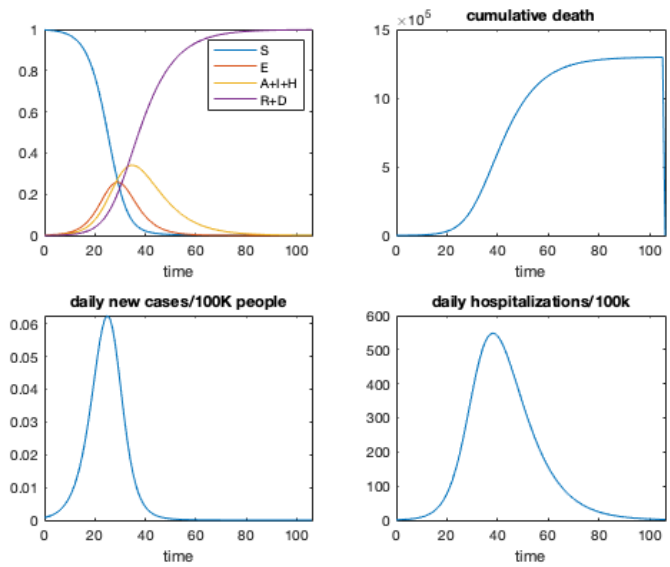
## Contents


- ii
- b
- ii
- 3
- 4
- 5
- COMMON DATA
- INITIALIZATION
- COMPUTATION
- FUNCTION FOR rk4
- FUNCTION FOR THE DIFFERENTIAL EQUATION
- END


```
% Sherry Yang
% hw 5


% a. when setting pc = 0

%i
maxdays = 1000;
delta = 1;
pc = 0;
pca = 0;
R0 = 5.7;
h0 = 1;
r0 = 0;
popn = 325*1000000;

[S,E,A,I,H,R,D,lambda]=covid19_model(delta,pc,pca,R0,h0,r0);
t = 0:length(S)-1;
figure(1)
subplot(2,2,1)
plot(t,S)
hold on
plot(t,E)
plot(t,A+I+H)
plot(t,R+D)
legend('S','E','A+I+H','R+D')
xlabel('time')

cum_death = zeros(length(t),1);
for i = 1:length(t)-1
    cum_death(i) = D(i)*popn;
end

subplot(2,2,2)
plot(t, cum_death)
title('cumulative death')
xlabel('time')

daily_new = zeros(length(t),1);
for i = 1:length(t)-1
    daily_new(i) = S(i) - S(i+1);
end
subplot(2,2,3)
plot(t, daily_new)
title('daily new cases/100K people')
xlabel('time')


subplot(2,2,4)
plot(t,H*100000)
title('daily hospitalizations/100k')
xlabel('time')
hold off
```



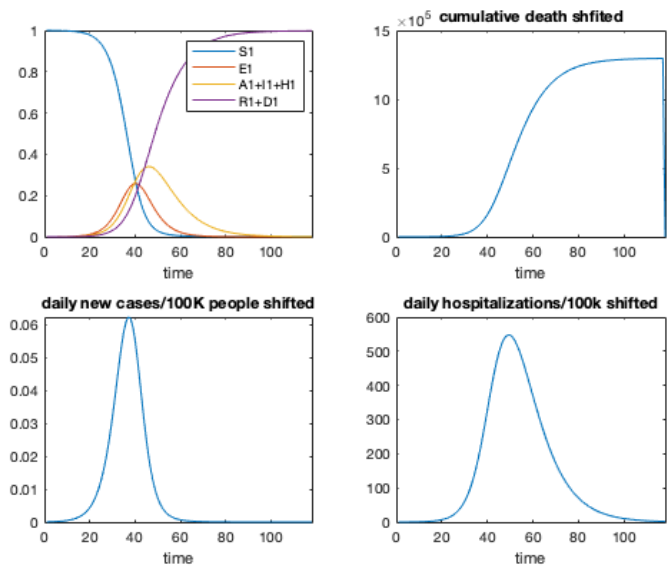

### ii

```
max_hos = max(H) * 10000;
%maximum number of patients requireing hospitalization per 100k people is
%548

% iii
max_y_indices = find(H == max(H)); % 39
% time point 39 is when hospitalization requiremnt reaches maximum


% iv
sum = zeros(length(t),1);
for i = 1:length(t)

    sum(i) = A(i) + I(i) + H(i);

    if i > 57 && sum(i) < 0.001
        break;
    end
end
% after plotting, the maximum is in the first half of all time point
% on the 103 time point,
% infected popn dropped below 100/100k , which is around 0.0001, outbreak
% ended


% v
infected = S + E;
infected_rate = 100-infected*100;
% percentage of popn that gets the disease during simulation: is 99.85%


%vi
total_death_1 = D(length(t)) * popn;
% vi total popn that dies is the last time point of graph 2,2,2, which is
% 1.3 million people
```

### b

```
h1 = 0.1;
t1 = linspace(1,1000,118);
%b
[S1,E1,A1,I1,H1,R1,D1,lambda1]=covid19_model(delta,pc,pca,R0,h1,r0);

figure(2)
subplot(2,2,1)
plot(1:length(S1),S1)
hold on
plot(1:length(E1),E1)
plot(1:length(A1),A1+I1+H1)
plot(1:length(R1),R1+D1)
legend('S1','E1','A1+I1+H1','R1+D1')
```

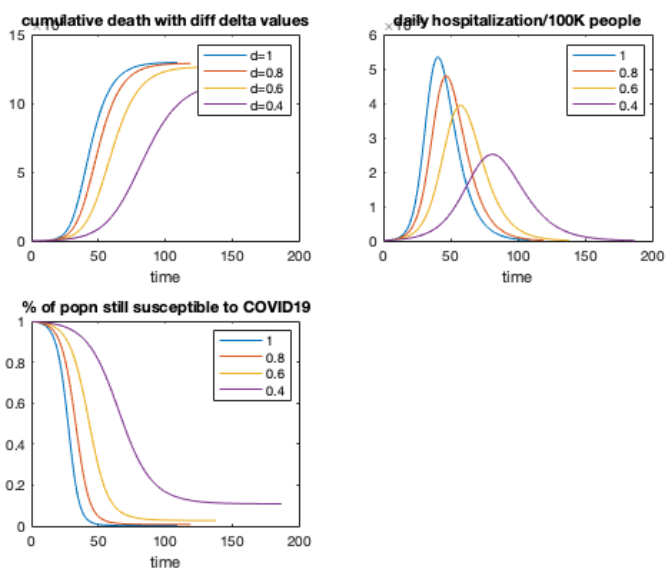
```
xlabel('time')
% comparing this graph to graph from part 1, the time for A+I+H to reach
% peak is 11 days later , although the peak number is very similar. the whole graph shifted rightward significantly
% a lot of people also could've not been dead (also shifted about 50 days),
% as well as daily new cases

cum_death1 = zeros(length(t1),1);
for i = 1:length(D1)-1
    cum_death1(i) = D1(i) * popn;
end


subplot(2,2,2)
plot(1:length(D1), cum_death1)
title('cumulative death shfited')
xlabel('time')

daily_new1 = zeros(length(t1),1);
for i = 1:length(S1)-1
    daily_new1(i+1) = S1(i) - S1(i+1);
end
subplot(2,2,3)
plot(1:length(S1), daily_new1)
title('daily new cases/100K people shifted')
xlabel('time')


subplot(2,2,4)
plot(1:length(H1),H1*100000)
title('daily hospitalizations/100k shifted')
xlabel('time')
hold off
```




2 a

```
pc1 = 0.1;
delta = [1 0.8 0.6 0.4];


[S2,E2,A2,I2,H2,R2,D2,lambda2]=covid19_model(delta(1),pc1,pca,R0,h0,r0);
[S3,E3,A3,I3,H3,R3,D3,lambda3]=covid19_model(delta(2),pc1,pca,R0,h0,r0);
[S4,E4,A4,I4,H4,R4,D4,lambda4]=covid19_model(delta(3),pc1,pca,R0,h0,r0);
[S5,E5,A5,I5,H5,R5,D5,lambda5]=covid19_model(delta(4),pc1,pca,R0,h0,r0);

death_d1 = cumulative_death(D2,popn);
death_d08 = cumulative_death(D3,popn);
death_d06 = cumulative_death(D4,popn);
death_d04 = cumulative_death(D5,popn);

figure(3)
subplot(2,2,1)
dd = {death_d1 death_d08 death_d06 death_d04};
for i = 1:4
```

```
    plot(1:length(dd{i}),dd{i})
    hold on
end
title('cumulative death with diff delta values')
legend('d=1','d=0.8','d=0.6','d=0.4')
xlabel('time')


subplot(2,2,2)
hh = {H2,H3,H4,H5};
for i = 1:4
    plot(1:length(hh{i}),hh{i})
    hold on
end
legend('1','0.8','0.6','0.4')
xlabel('time')
title('daily hospitalization/100K people')


subplot(2,2,3)
ss = {S2,S3,S4,S5};
for i = 1:4
    plot(1:length(ss{i}),ss{i})
    hold on
end
legend('1','0.8','0.6','0.4')
xlabel('time')
title('% of popn still susceptible to COVID19')




% b
```




## ii

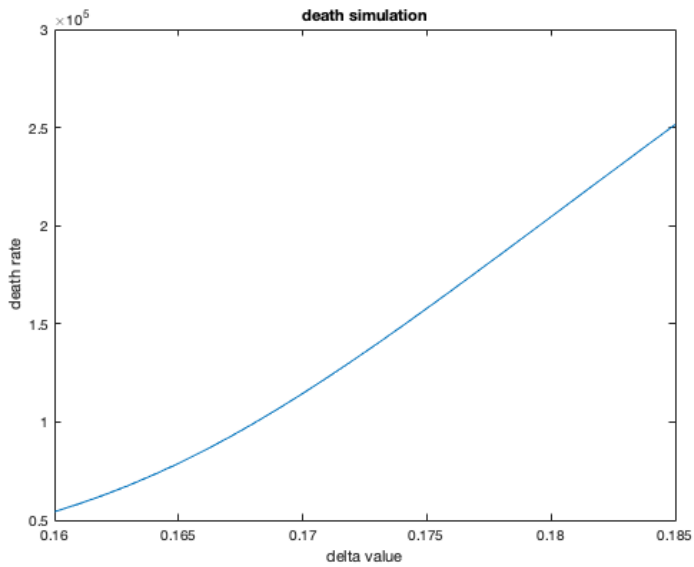
```
% i
[max_hos_vec] = [max(H2) max(H3) max(H4) max(H5)]* 100000;
%maximum number of patients requireing hospitalization per 100k people is
%534,479,393,251

% ii
max_y_indices_H2 = find(H2 == max(H2));
max_y_indices_H3 = find(H3 == max(H3));
max_y_indices_H4 = find(H4 == max(H4));
max_y_indices_H5 = find(H5 == max(H5));
% the day on hich ospitalization requireement reaches its maximum: 41 47 57 81

% iii
A = {A2 A3 A4 A5};
I = {I2 I3 I4 I5};
H = {H2 H3 H4 H5};
S = {S2 S3 S4 S5};
E = {E2 E3 E4 E5};
```

```
total_infected = {};

stop_j = zeros(1, 4);
for i = 1:4
    total_infected{i} = A{i} + I{i} + H{i};
    max_day(i) = find(total_infected{i}==max(total_infected{i}));
    for j = max_day(i):length(total_infected{i})
        if total_infected{i}(j)<0.001 ;
            stop_j(i) = j;
            break
        end
    end
end

% end of out break are 104, 113, 131, 176 time points for d = 1,.8,.6,.4

%iv
infected_vec = {};
infected_rate_vec = {};
for i = 1:4
    infected_vec{i} = S{i} + E{i};
    infected_rate_vec{i} = 100-infected_vec{i}*100;
end
%percentage of popn that gets the disease during simulation: 99.78%, 99.24%,97.29%, 89.15%

% v
for i = 1:4
    total_death(i) = dd{i}(length(dd{i}));
end

% total death are 1.3, 1.29, 1.26, 1.16 million for d = 1,.8,.6,.4
```

## 3

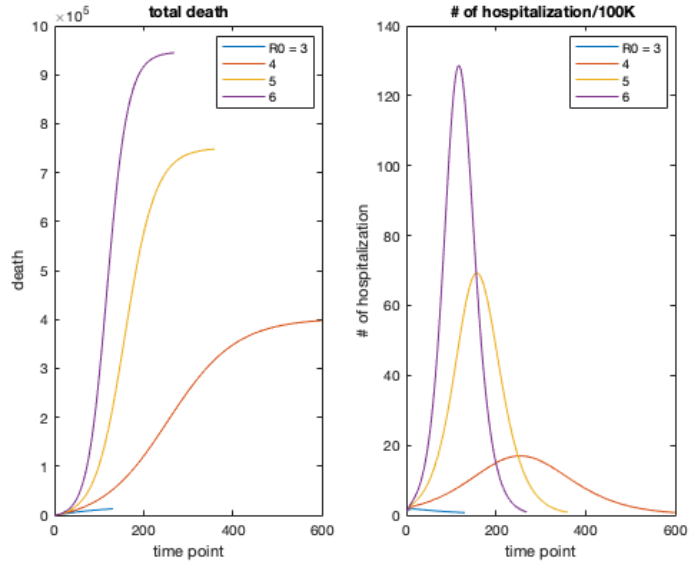
```
pc3 = 0.3;
delta = [.24,.22,.2,.18,.16];
for i = 1:length(delta)
    [S_total{i},E_total{i},A_total{i},I_total{i},H_total{i},R_total{i},D_total{i},lambda_total{i}]=covid19_model(delta(i),pc3,pca,R0,h0,r0);
end


% i
for i = 1:5
    max_hos_vector{i} = max(H_total{i})* 100000;
end

%maximum number of patients requireing hospitalization per 100k people is
%59, 32, 16, 6, 3

% ii
for i = 1:5
    max_day_indices{i} = find(H_total{i} == max(H_total{i}));
end
% the day on which wospitalization requireement reaches its maximum: 175,
% 202,238,239,21

% iii

for i = 1:5
    total_infected_low{i} = A_total{i} + I_total{i} + H_total{i};
    max_day_low(i) = find(total_infected_low{i}==max(total_infected_low{i}));
    for j = max_day_low(i):length(total_infected_low{i})
        if total_infected_low{i}(j)<0.001 ;
            stop_j_low(i) = j;
            break
        end
    end
end

% end of out break are 370, 431,515,567,183 time points for d = .24,.22,.2,.18,.16


%iv
for i = 1:5
    infected_total{i} = S_total{i} + E_total{i};
    infected_rate_total{i} = 100-infected_total{i}*100;
end
% % of popn that gets disease during simu: 50.19%, 40.96%, 29.54%, 15%, 4.2%
```

```
% v

for i = 1:5
    total_death_low(i) = D_total{i}(length(D_total{i}))*popn;
end

% total death are 65, 53, 38, 20, and 54 K for d = .24,.22,.2,.18,.16


%b
% yes its definitely very effective when practcing lower
% contact/transmission rate. the time of max hospitalization and end of out
% break are much more delayed, but the death rate and max number of people
% requireing hsopitaization are much much lower, saving many lives. it
% blunts the outbreak by roughly 100 days, about 5000 times less people died



figure(4)
l = total_death_low*popn;
% results are 6.5211    5.3199    3.8364    2.0453    0.5427 10e5
plot(delta,l)
xlabel('delta values/aggressive measures level')
ylabel('total number of death')
title('effectiveness of aggressive measures')
% according to the plot theres somewhat a linear relationship between
% agreesive measures and total numebr of death. his claim is roughly correct with
% most extreme measures, it will be close to 100 to 200 k


%d

sim = 0.16:0.001:0.185;
death_sim = zeros(26,1);
for i = 1:length(sim)

    [S_sim{i},E_sim{i},A_sim{i},I_sim{i},H_sim{i},R_sim{i},D_sim{i},lambda_sim{i}]=covid19_model(sim(i),pc3,pca,R0,h0,r0);
    death_sim(i) = D_sim{i}(length(D_sim{i}));
end
death_sim_popn = death_sim * popn;
figure(5)
plot(sim,death_sim_popn)
title('death simulation')
xlabel('delta value')
ylabel('death rate')

lb_d = interp1(death_sim_popn, sim, 100000);
ub_d = interp1(death_sim_popn, sim, 200000);


% according to figure 5, when d is between 0.168 and 0.180, death is
% between 100 to 200K


% e with both aggresive measure of masks and physical distance, it
% decreases delta by a factor of 8, the model does support that more
% aggressive measure wold have signifiacntly impacted the death rate. When:
% delta = 1, death rate = 1.3 million
% delta rate = 0.168, death rate is 0.1 million
% with 5 fold more aggeressiv emeasures, death raet is lowered by 13 fold
% the peak is delayed a lot, final susceptible percentage decrease
% signifincantly
```




## 4

```
delta_03 = 0.3;
r0_vec = [3,4,5,6];
for i = 1:4

    [S_vec{i},E_vec{i},A_vec{i},I_vec{i},H_vec{i},R_vec{i},D_vec{i},lambda_vec{i}]=covid19_model(delta_03,pc3,pca,r0_vec(i),h0,r0);
    daily_changes{i} = diff(H_vec{i});
    daily_growth_rate{i} = (daily_changes{i}(2:end) - daily_changes{i}(1:end-1)) ./ daily_changes{i}(1:end-1) ;
end

figure(6)
subplot(1,2,1)
for i = 1:4
    plot(1:length(D_vec{i}),D_vec{i}*popn)
    hold on
end
legend('R0 = 3','4','5','6')
ylabel('death')
xlabel('time point')
title('total death')

subplot(1,2,2)
for i = 1:4
    plot(1:length(H_vec{i}),H_vec{i}*100000)
    hold on
end
legend('R0 = 3','4','5','6')
ylabel('# of hospitalization')
xlabel('time point')
title(' # of hospitalization/100K')
% both total death and rate of increase in hospitalization has signifiacant
% fluctuation with different values of R0. its hard to see a stable trend
% in both graphs, the larger the R0, the more death
```




## 5

```
r0_sim = 4*rand(5000, 1) + 2;

for i = 1:length(r0_sim)
    [S_vec1{i},E_vec1{i},A_vec1{i},I_vec1{i},H_vec1{i},R_vec1{i},D_vec1{i},lambda_vec1{i}]=covid19_model(delta_03,pc3,pca,r0_sim(i),h0,r0);
end

figure(7)
for i = 1:length(r0_sim)
    plot(1:length(D_vec1{i}),D_vec1{i}*popn)
    hold on
end
xlabel('time')
ylabel('# of death')
title(' # death over time for 5000 simulation')

% b

for i = 1:length(D_vec1)
    D_last(i) = D_vec1{i}(length(D_vec1{i}))*popn;
end


sum_mean = 0;
for i = 1:length(D_last)-1

    sum(i) = D_last(i);
    sum_mean = (sum_mean + sum(i));
    mean_vec(i) = sum_mean/i;
end
figure(8)
plot(1:length(mean_vec),mean_vec)
ylabel('mean death population')
xlabel('# of simulation')
title('mean death population over 5000 simulations')

%c
for i = 1:length(D_vec1)-1
    sd(i) = std(mean_vec(1:i));
end

figure(9)
plot(1:length(D_vec1)-1,sd)
ylabel('sd')
xlabel('# of simulation')
title('sd over 5000 simulations')
hold off

%d
for i = 1:length(D_vec1)-1
    true_mean = mean_vec(length(D_vec)-1);
    absolute_error(i) = abs(mean_vec(i) - true_mean);
    percent_error(i) = (absolute_error(i) / true_mean);
end
```

```
figure(10)
plot(1:length(percent_error),percent_error)
ylabel('percentage error')
xlabel('# of simulation')
title('percentage error over 5000 simulation')
hold off
% after 2500 simulation, the error stabilizes around 0%, estimate of the
% mean stabilize around 40K


% e
sd1= [10 100 1000 5000];
for i = 1:4
    lb_mean(i) = true_mean - 2*sd1(i);
    ub_mean(i) = true_mean + 2*sd1(i);
end

% confidence intervals are as follows:
% lowerbound 3.68, 3.67, 3.66, 3.58 e5
% upperbound 3.68 3.68 3.70 3.78 e5

% f

realdata = readtable("covid_deaths.csv");
realdata = table2array(realdata);
figure(11)
plot(realdata(:,1),realdata(:,2))
hold on
B = repmat(true_mean,766);
plot(1:766,B)
legend('real data','calculated')
ylabel('death rate')
title('real data vs. computed mean')

% the real data is much higher than estimate mean, its not within the confidence
% interval. but depending on random simulation, the value actually
% fluctuates a lot.



function [death] = cumulative_death(d,popn)
    death = zeros(length(d),1);
    for i = 1:length(d)
        death(i) = d(i)*popn;
    end
end

function [S,E,A,I,H,R,D,lambda]=covid19_model(delta,pc,pca,R0,h0,r0)
```

```
% function [S,E,A,I,H,R,D,lambda]=covid19_sim(delta,pc,pca,R0,h0,r0)
%
% runs a simulation of an SEAIHDR model
%
% S: susceptible
% E: exposed
% A: asymptomatic
% I: infective (symptomatic)
% H: hospitalized
% R: recovered
% D: deceased
% lambda: initial exponential growth rate
%
% delta is the fractional contact rate for social distancing
% pa is the fraction of infectives who are asymptomatic
% pc is the fraction of symptomatics who are tested
% pca is the fraction of asymptomatics who are tested
% R0 is the basic reproductive number with no detection or intervention
% ph is the fraction of infectives who require hospitalization
% mh is the fraction of hospitalized patients who die
% fa, fc, fh are the infectivities of A, C, and H, relative to that of I
% te, ta, ti, tc, th are the expected durations of classes E, A, I1, I2, H
% h0 is the number of initial hospitalized infectives per hundred thousand
% r0 is the fraction of the population that is initially immune
% target is the value of A+I+H per hundred thousand defined as the end condition
%
% results is a matrix: columns are SEAI1I2HRD, rows are 0:days
```

```
%
% adapted from UNL
```

## COMMON DATA

```
pa = 0.4;
ph = 0.016;
mh = 0.25;
fa = 0.75;
fc = 0.1;
fh = 0.05;
te = 5;
ta = 8;
ti = 10;
tc = 3;
th = 10;
target = 60;

maxdays = 1000;
```

## INITIALIZATION

```
% set derived parameters

h0 = h0/100000;
target = target/100000;

eta = 1/te;
alpha = 1/ta;
gamma = 1/ti;
nu = 1/th;
sigma = 1/tc;

beta = R0/(fa*pa*ta+(1-pa)*ta);

M = [-eta,fa*beta,beta,beta,fh*beta;pa*eta,-alpha,0,0,0;(1-pa-ph)*eta,0,-gamma,0,0;ph*eta,0,0,-sigma,0;0,0,0,sigma,-nu];
lambda = max(eig(M));

i20 = h0*(lambda+nu)/sigma;
e0 = i20*(lambda+sigma)/(ph*eta);
a0 = e0*pa*eta/(lambda+alpha);
i10 = e0*(1-pa-ph)*eta/(lambda+alpha);

% set up results data structure with Y=[S,E,A,I1,I2,H,R,D]

results = zeros(maxdays+1,8);
Y = [0,e0,a0,i10,i20,h0,r0,0];
Y(1) = 1-sum(Y);
results(1,:) = Y;

y = Y';
summ = e0+a0+i10+i20+h0;   % summ is used to help determine the end condition
```

## COMPUTATION

```
for t=1:maxdays
    % y is a column vector, Y^T
    y = rk4(1,y);
    Y = y';
    results(t+1,:) = Y;
    if sum(Y(2:6))>min(target,summ)
        summ = sum(Y(2:6));
    else
        results = results(1:(t+1),:);
        break;
    end
end

S = results(:,1);
E = results(:,2);
A = results(:,3);
I = results(:,4)+results(:,5);
H = results(:,6);
R = results(:,7);
D = results(:,8);
```

## FUNCTION FOR rk4

```
    function y=rk4(dt,y0)
        % y0 is a column vector of initial conditions at t
        % y is a column vector of values at t+dt
        k1 = yprime(y0);
        k2 = yprime(y0+0.5*dt*k1);
        k3 = yprime(y0+0.5*dt*k2);
        k4 = yprime(y0+dt*k3);
        y = y0+dt*(k1+2*k2+2*k3+k4)/6;
    end
```

## FUNCTION FOR THE DIFFERENTIAL EQUATION

```
    function yp=yprime(y)
    % split out components
        S = y(1);
        E = y(2);
        A = y(3);
        I1 = y(4);
        I2 = y(5);
        I = I1+I2;
        H = y(6);
        X = delta*(fa*(1-pca)*A+(1-pc)*I)+fc*(pca*A+pc*I)+fh*H;
    % compute derivatives
        Sp = -beta*X*S;
        Ep = -Sp-eta*E;
        Ap = pa*eta*E-alpha*A;
        I1p = (1-pa-ph)*eta*E-gamma*I1;
```

```
        I2p = ph*eta*E-sigma*I2;
        Hp = sigma*I2-nu*H;
        Rp = alpha*A+gamma*I1+(1-mh)*nu*H;
        Dp = mh*nu*H;
    % assemble derivative
        yp = [Sp;Ep;Ap;I1p;I2p;Hp;Rp;Dp];
    end
```

## END

```
end
```

```
Warning: Column headers from the file were modified to make them valid MATLAB
identifiers before creating variable names for the table. The original column
headers are saved in the VariableDescriptions property.
Set 'VariableNamingRule' to 'preserve' to use the original column headers as
table variable names.
```

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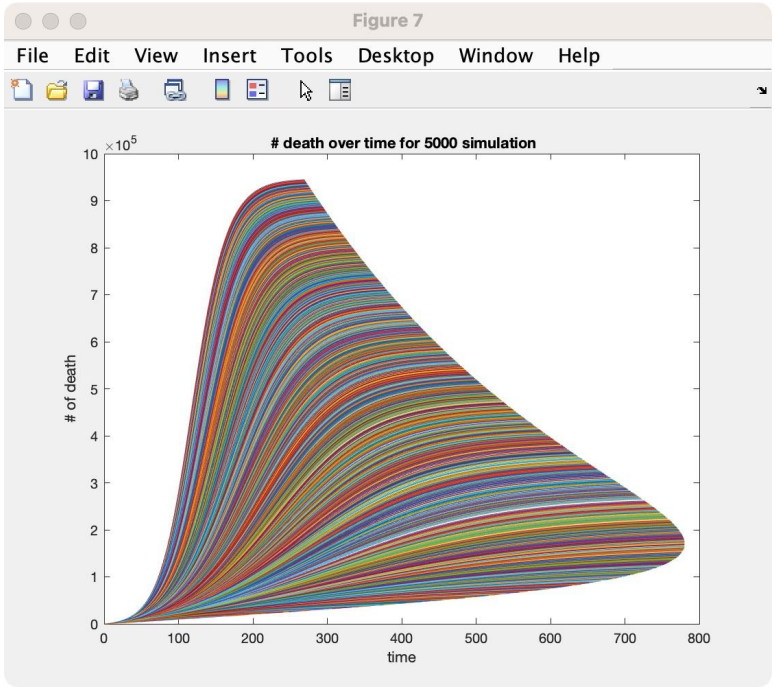
Figure 7
File Edit View Insert Tools Desktop Window Help
# death over time for 5000 simulation
×10^5
# of death
time
0
1
2
3
4
5
6
7
8
9
10
0
100
200
300
400
500
600
700
800

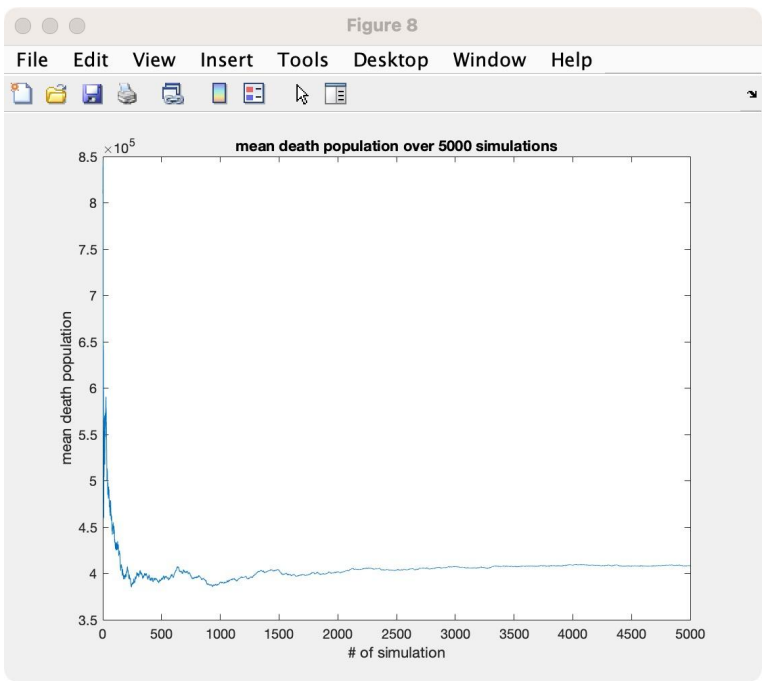
Figure 8
File Edit View Insert Tools Desktop Window Help
mean death population over 5000 simulations
×10^5
mean death population
# of simulation
3.5
4
4.5
5
5.5
6
6.5
7
7.5
8
8.5
0
500
1000
1500
2000
2500
3000
3500
4000
4500
5000

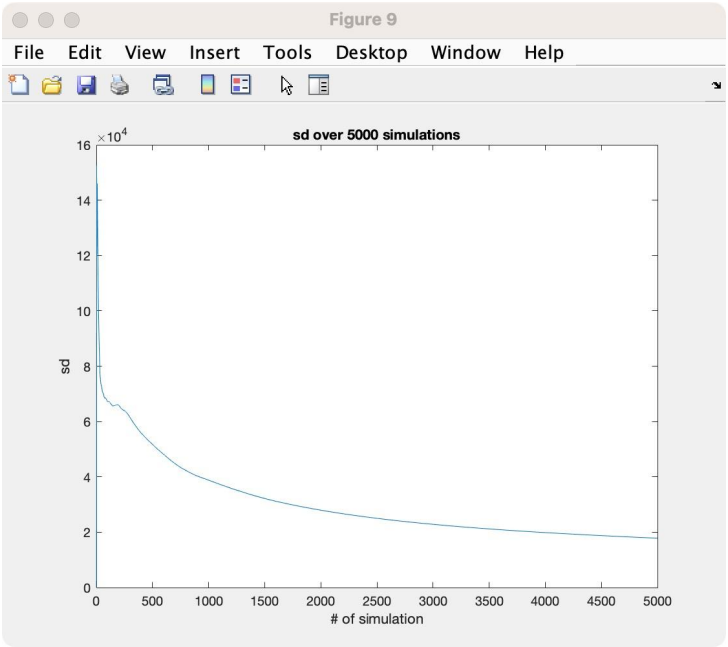
Figure 9
File Edit View Insert Tools Desktop Window Help
sd over 5000 simulations
×10^4
16
14
12
10
8
6
4
2
0
sd
0 500 1000 1500 2000 2500 3000 3500 4000 4500 5000
# of simulation

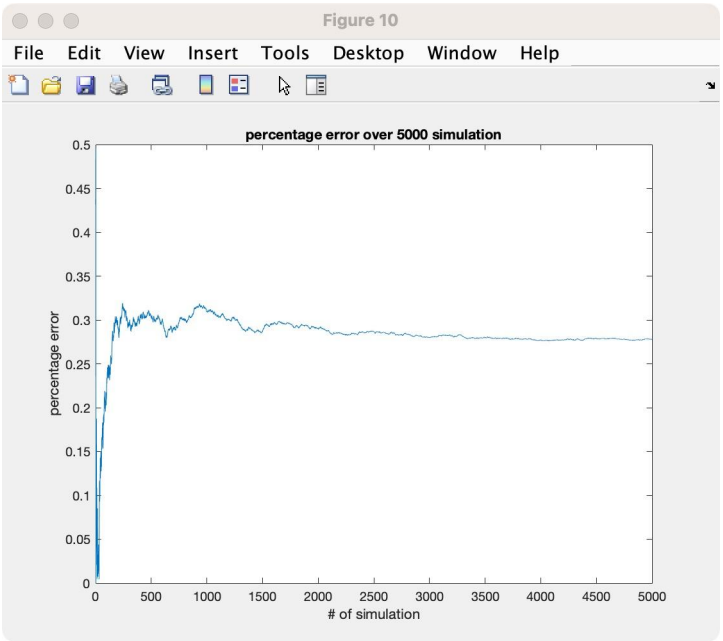
Figure 10
File Edit View Insert Tools Desktop Window Help
percentage error over 5000 simulation
0.5
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
percentage error
0 500 1000 1500 2000 2500 3000 3500 4000 4500 5000
# of simulation

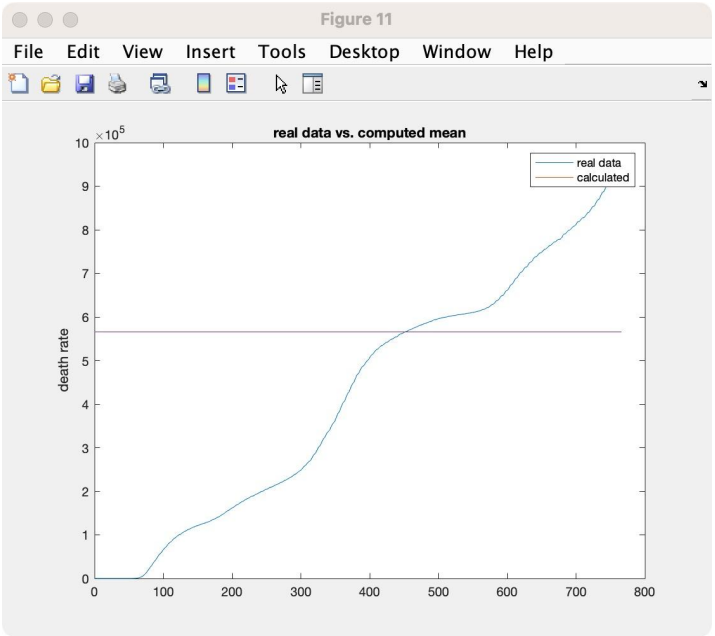
Figure 11
File Edit View Insert Tools Desktop Window Help
real data vs. computed mean
×10^5
death rate
real data
calculated
0
1
2
3
4
5
6
7
8
9
10
0
100
200
300
400
500
600
700
800